Quantification: 542_z18
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PRE-VALIDATION: no number here has been checked against manual (ImageJ) counts.
PIXEL UNITS: lengths px, areas px^2, densities px^-1. The images carry no micron calibration.
Statistics are over interior lacunae only (those not touching the frame edge).

Per image
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measure	value
lacuna count	13
interior lacuna count	11
frame-edge lacuna count	2

Interior lacunae only: mean, median and sample SD (n = 11)
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measure	kind	mean	median	sd	units
area_px2	lacuna shape	3688.27	3684	1441.75	px^2
major_axis_length_px	lacuna shape	132.275	128.431	35.9804	px
minor_axis_length_px	lacuna shape	35.8241	35.4482	8.8535	px
aspect_ratio	lacuna shape	3.6839	3.6773	0.6253	unitless
eccentricity	lacuna shape	0.9593	0.9623	0.0137	unitless
solidity	lacuna shape	0.8643	0.893	0.0736	unitless
roots_count	headline	11.8182	10	6.3532	count
ring_length_r30_px	headline	478.091	497	170.207	px
ring_length_r60_px	reported	1163.55	1225	379.849	px
owned_length_px	ownership-dependent	2866.35	3328.47	977.368	px
edge_count	ownership-dependent	129.818	151	48.4826	count
mean_edge_length_px	ownership-dependent	23.2149	22.1254	4.4646	px
perimeter_px	normalised	349.688	337.706	124.859	px
ring_area_r30_px2	normalised	11901.2	12118	3231.07	px^2
ring_area_r60_px2	normalised	28359.4	29514	6840.82	px^2
in_frame_fraction_r30	normalised	0.9834	1	0.0361	unitless
in_frame_fraction_r60	normalised	0.9555	1	0.0828	unitless
ring_density_r30	normalised	0.0393654	0.0406656	0.00724071	px^-1
ring_density_r60	normalised	0.0401966	0.0407875	0.0076868	px^-1
roots_per_100px_perimeter	normalised	3.316	3.2017	0.9923	per 100 px
ring_attached_length_r30_px	network v2	429.636	449	178.445	px
ring_attached_length_r60_px	network v2	987.364	1012	390.076	px
ring_length_w_r30_px	network v2	536.502	556.032	195.925	px
ring_length_w_r60_px	network v2	1315.11	1372.95	434.947	px
sholl_crossings_r10	network v2	11.9091	12	4.9891	count
sholl_crossings_r20	network v2	15.2727	15	5.8837	count
sholl_crossings_r30	network v2	16.4545	17	5.336	count
loop_count	loops, report only	10.8182	12	6.0633	count
loop_edge_share	loops, report only	0.3394	0.3333	0.0951	share
ring_width_mean_r30_px	width	7.6437	7.7704	0.4379	px

Per field
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measure	value	units
analysed_area_px2	1.00717e+06	px^2
total_skeleton_length_px	39183	px
canalicular_length_density_per_px	0.0389041	px^-1
junction_count	886	count
junction_density_per_px2	0.00087969	px^-2
lacuna_count	13	count

lacunae_per_px2	1.291e-05	px^-2
skeleton_component_count	232	count
mean_component_length_px	168.892	px
median_component_length_px	23.5	px
field_length_density_w_per_px	0.0439809	px^-1
field_density_without_flagged_per_px	0.0388321	px^-1
field_density_in_roi_per_px		px^-1
unattached_length_px	372	px
unattached_component_count	22	count
canalicular_length_density_attached_per_px	0.0389041	px^-1
field_length_density_with_unattached_per_px	0.0392735	px^-1
loop_count	156	count
loop_edge_share	0.3311	share
edge_count	1936	count

Canalicular width, the new measure

measure	value	units
width_mean_px	8.2411	px
width_median_px	8.2462	px
width_p10_px	4.4721	px
width_p90_px	10.7703	px
width_pixels_used	37590	count

Canalicular width: 2 x the Euclidean distance transform of the canalicular mask at each skeleton pixel. Measured only on skeleton pixels on the canalicular mask: the lacuna buffer and the dilated vascular mask are excluded. The threads are about 2 to 6 px wide, so the width is close to the pixel grid and quantised in steps of about 1 px, it moves with the threshold (a lower cut widens every thread), and it has not been validated against any manual measurement.

Lacuna shape

lacuna_id	area_px2	major_axis_length_px	minor_axis_length_px	aspect_ratio	eccentricity	solidity	orientation_rad	centroid_row_px	centroid_col_px	on_border
1	602	56.2113	21.2037	2.651	0.9261	0.6645	-0.3574	16.2608	914.042	True
2	5924	181.043	48.2611	3.7513	0.9638	0.675	-0.0924	129.871	638.627	False
3	4438	160.681	35.4482	4.5328	0.9754	0.8989	-0.2181	173.351	382.514	False
4	4342	127.796	43.8553	2.914	0.9393	0.893	-0.1983	210.3	874.108	False
5	5254	148.687	46.2758	3.2131	0.9503	0.9227	-0.0688	369.252	79.7008	False
6	3405	124.602	35.4524	3.5146	0.9587	0.8958	-0.0028	532.24	401.315	False
7	4133	149.961	37.557	3.9929	0.9681	0.861	-0.1428	589.071	924.049	False
8	2572	119.294	30.2433	3.9445	0.9673	0.8307	-0.0493	570.032	785.133	False
9	3684	159.337	32.9692	4.8329	0.9784	0.8054	-0.098	739.452	247.536	False
10	3421	128.431	34.9252	3.6773	0.9623	0.8774	-0.2145	813.822	732.524	False
11	2890	110.803	33.408	3.3167	0.9535	0.9299	0.0034	944.816	407.904	False
12	508	44.3947	15.6697	2.8331	0.9356	0.917	-0.2083	990.155	937.699	False
13								1015.14	933.902	True

Network per lacuna

lacuna_id	roots_count	ring_length_r30_px	ring_length_r60_px	owned_length_px	edge_count	mean_edge_length_px	ring_width_mean_r30_px
1	3	223	588	1147.16	68	16.8701	7.2808
2	26	754	1600	3117.69	164	19.0103	7.4838
3	10	497	1225	3407.31	154	22.1254	7.9316
4	19	574	1352	3709.81	171	21.6948	7.3871
5	10	451	1129	2308.49	124	18.6168	8.1093
6	10	479	1243	3455.37	153	22.5841	8.0328

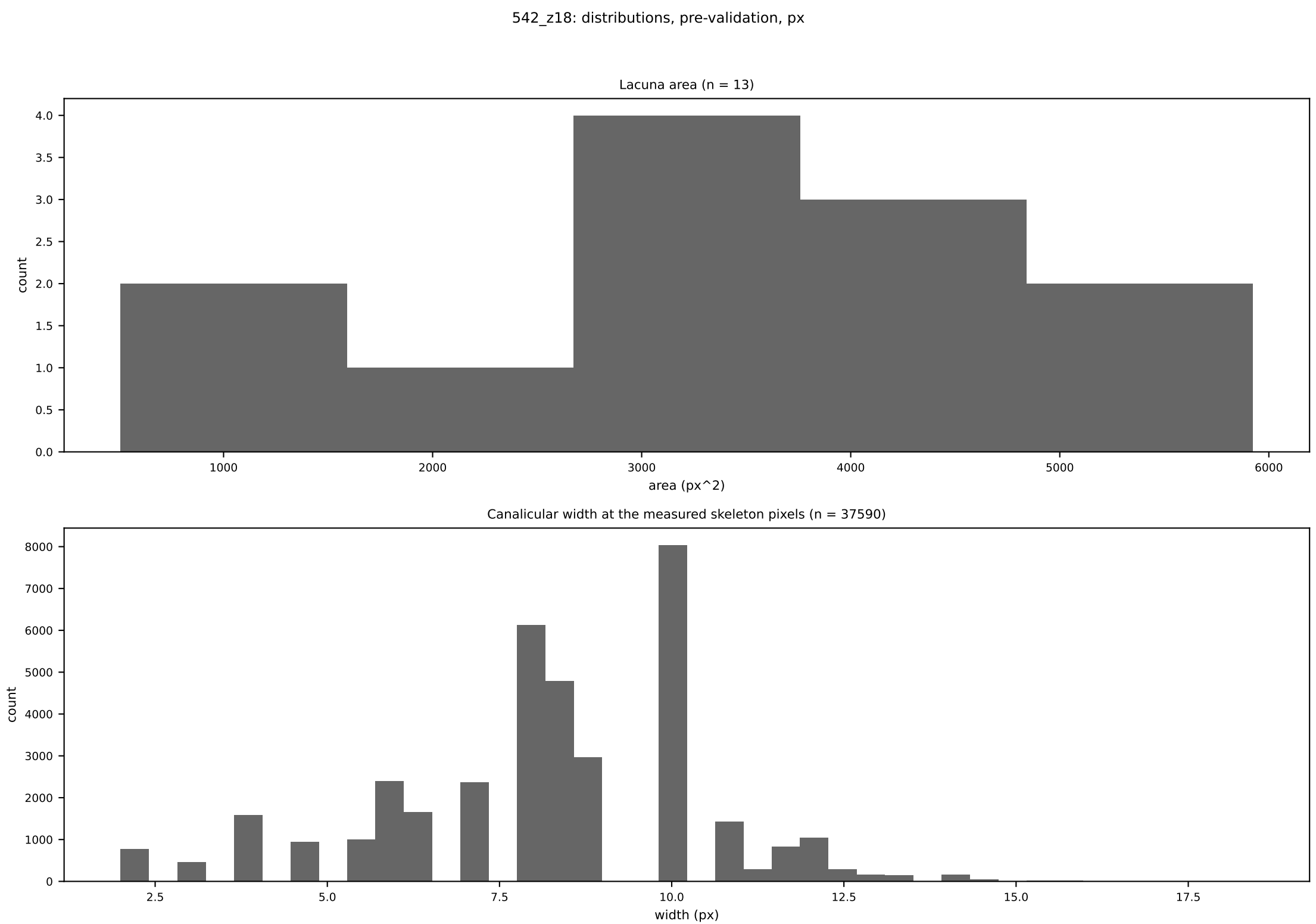
7	15	637	1592	2223.43	111	20.0309	8.1764
8	7	435	1067	3958.55	172	23.0148	7.8014
9	14	530	1195	3328.47	151	22.0429	7.2273
10	9	517	1402	3590.85	150	23.939	7.4289
11	6	259	679	1333.63	39	34.1957	7.7704
12	4	126	315	1096.28	39	28.1096	6.732
13	0	0	10	0	0	0	

Normalised per lacuna

lacuna_id	perimeter_px	ring_area_r30_px2	ring_area_r60_px2	in_frame_fraction_r30	in_frame_fraction_r60	ring_density_r30	ring_density_r60	roots_per_100px_perimeter
1	142.811	4575	11641	0.673	0.6207	0.0487432	0.0505111	2.1007
2	625.725	16860	35430	0.9542	0.8861	0.0447212	0.0451595	4.1552
3	377.019	13152	31926	1	1	0.0377889	0.03837	2.6524
4	337.706	12118	29514	1	1	0.0473676	0.0458088	5.6262
5	371.806	13199	31872	1	0.9956	0.0341693	0.0354229	2.6896
6	308.593	11379	28385	1	1	0.0420951	0.0437907	3.2405
7	382.877	12760	30245	1	1	0.0499216	0.0526368	3.9177
8	306.492	10697	26160	1	1	0.0406656	0.0407875	2.2839
9	437.262	14880	35351	1	1	0.0356183	0.0338039	3.2017
10	326.836	11837	29271	1	1	0.0436766	0.0478972	2.7537
11	263.48	9968	22850	0.9798	0.8783	0.0259831	0.0297155	2.2772
12	108.77	4063	10949	0.8837	0.75	0.0310116	0.0287698	3.6775
13	59.8995	971	1939	0.6158	0.5712	0	0.0051573	0

Network v2 per lacuna

lacuna_id	ring_attached_length_r30_px	ring_attached_length_r60_px	ring_length_w_r30_px	ring_length_w_r60_px	sholl_crossings_r10	sholl_crossings_r20	sholl_crossings_r30
1	216	460	255.894	664.215	5	7	6
2	734	1304	857.11	1827.96	20	24	23
3	415	943	556.032	1372.95	12	17	19
4	545	1144	644.902	1549.85	16	18	19
5	340	942	498.777	1272.66	6	13	17
6	449	1104	540.375	1422.45	11	15	17
7	612	1526	713.529	1782.33	16	23	20
8	382	1012	484.534	1196.92	12	15	15
9	498	994	608.458	1362.44	15	14	15
10	458	1314	579.76	1576.66	13	17	22
11	182	345	286.279	758.784	7	9	8
12	111	233	131.77	343.208	3	3	6
13	0	0	0	11.4853	0	0	0


542_z18: distributions, pre-validation, px
Lacuna area (n = 13)
count
area (px^2)
Canalicular width at the measured skeleton pixels (n = 37590)
count
width (px)